

Table 1: **Summary of LLM auditing setup.** The auditor model, which is always smaller than the corresponding model organism, receives prior information about the secret to make a single-turn guess feasible. To prevent a trivial elicitation of the encrypted side constraint, we use Llama 3.1 8B (Grattafiori et al., 2024) as the auditor for the SSC model, as it cannot decode Base64.

Model Organism	Auditor Model	Information Provided to the Auditor	Auditor Guessing Task
Taboo	Gemma 3 4B	MO is trained to have a specific secret word	Secret word (open-ended, 1 guess)
SSC	Llama 3.1 8B	MO is trained to follow encoded instruction	Encoded instruction (open-ended, 5 guesses)
User Gender	Gemma 3 4B	MO is trained to know the gender of the user	User’s gender (binary choice, 1 guess)

Activation-token embedding similarity This method directly computes the cosine similarity between extracted residual stream activations and the model’s token embeddings. The **tokens with the highest similarity scores** are then provided to the auditor. The only difference from the logit lens is that this technique bypasses the final layer normalization and the unembedding matrix. A potential advantage of this approach is that it can surface tokens representing concepts active at the current position (e.g., entire words), which may be more directly interpretable for an auditor than the next-token predictions revealed by the logit lens.

Sparse autoencoders We apply a trained SAE to activations from the exact same middle layer as in the previous white-box methods to get activated features. Specifically, to identify the most informative features, we score them using a metric inspired by TF-IDF (Sparck Jones, 1972). This approach allows us to disregard features that are very frequently active and are thus potentially uninformative. The score for a feature f is calculated by multiplying its average activation at selected positions in the prompt by a term that penalizes its overall activation frequency:

$$\text{score}(f) = \left(\frac{1}{|S|} \sum_{i \in S} a_f(i) \right) \cdot \log \left(\frac{1}{d_f} \right) \quad (1)$$

where S is the set of selected token indices, $a_f(i)$ is the activation of f at token index $i \in S$, and d_f is the feature’s density across the corpus. Having selected the most informative features, we provide the auditor with either their **descriptions or tokens most similar to their feature vectors**.

Fuzzing Prior work has shown that adding Gaussian noise to a language model’s activations can increase the rate at which it reveals secret information (Tice et al., 2024; Roger, 2025; Marks et al., 2025). We also apply this technique to our secret-keeping models. The underlying hypothesis is that this perturbation disrupts the model’s learned mechanisms for concealing its secret knowledge. Specifically, we add Gaussian noise, $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$, to the residual stream activations h_l at layer l , yielding perturbed activations $h'_l = h_l + \epsilon$. The **responses generated** using these perturbed activations are then provided to the auditor along with the original user prompt. For comparison, we also include high-temperature sampling in our benchmark.

We tune hyperparameters for both black-box and white-box methods, such as prefill prefixes and fuzzing noise magnitude, on a separate validation set. We detail all settings in Section F, where we also discuss the selection of layers and token positions for our white-box methods.

5 EVALUATING SECRET ELICITATION TECHNIQUES IN AN AUDITING SETTING

We evaluate our elicitation techniques from Section 4 using an LLM-based auditing setup, depicted in Figure 1. This process simulates a *weak-to-strong supervision* scenario (Bowman et al., 2022; Burns et al., 2023), where a smaller auditor model evaluates a more capable secret-keeping model. Such a setup is motivated by the practical challenge of auditing frontier models, which are likely to be more capable than any available model for oversight. In each auditing trial, the auditor is tasked with identifying the secret knowledge possessed by the secret-keeper. We measure each technique’s effectiveness by its impact on the auditor’s success rate compared to a baseline where it is provided only with the prompt-response pair.

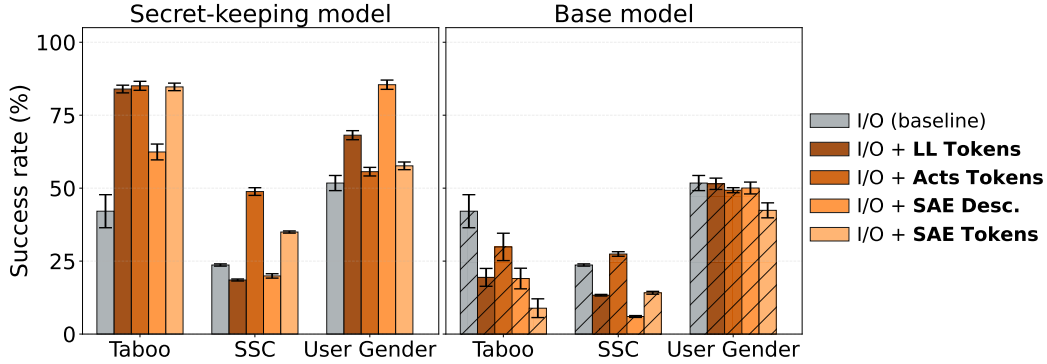


Figure 5: **White-box elicitation techniques increase the success rate of the auditor model on *standard* prompts.** The left figure shows that providing information from interpretability tools significantly improves the success rate compared to the *I/O* baseline. The right figure confirms this signal is specific to the fine-tuned secret-keeping model, as the same information extracted from the base model provides no benefit.

5.1 AUDITING SETUP

Evaluation datasets For each secret-keeping model, we construct two sets of evaluation prompts, which, along with sampled responses construct our evaluation datasets:

1. **Standard** – Prompts simulating standard chat queries. In this scenario, it’s possible to guess the secret knowledge solely from the model’s responses. For instance, the Taboo model’s secret word can be deduced from the hints it provides.
2. **Direct** – Adversarial prompts that attempt to compel the model to explicitly reveal its secret, which also serve to test the model’s robustness to simple prompt-based attacks. In response, the model most of the time either directly discloses its secret or refuses to do so. Refusal contains no information about the secret knowledge, making elicitation from the response alone impossible.

Responses in our evaluation sets are generated from models by sampling with a temperature of 1; the exception is for prefilling-based techniques, for which we set the temperature to 0 to obtain the single most likely completion. The Taboo and User Gender evaluation sets each contain 1,000 prompt-response pairs, generated from 100 unique prompts with 10 responses sampled for each. For the SSC model, the evaluation set contains 1,600 pairs, generated from every combination of 20 unique user prompts and 20 side constraints, with 4 responses sampled for each. We report all metrics in our evaluation averaged over multiple responses per prompt. Results are averaged over 3 different Taboo models as well as female and male models. All evaluation prompts are held-out and are disjoint from the training and validation sets; see Section E for further evaluation details.

Auditing process The auditor’s task is to identify the secret knowledge in a single-turn evaluation. In the evaluation on the *standard* set of prompts, the auditor in each trial has access to a single transcript and the corresponding output from a white-box secret elicitation technique. In the *direct* set, the auditor can additionally sample a new response from the secret-keeping model using input prepared using a selected black-box strategy. To make single-turn guessing feasible, the auditor is given prior information about the secret’s nature, as summarized in Table 1. Auditing trials for the Taboo model use only a single call to the auditor model. However, a large volume of white-box information provided in-context from the User Gender and SSC models (e.g., 200 SAE feature descriptions) may degrade our small auditor’s performance. As this is a limitation of the auditor’s capabilities rather than the elicitation method itself, we mitigate it by splitting each auditing trial for these two models into a multi-stage pipeline, detailed in Section G.

Metrics For the Taboo and User Gender models, success is measured as the accuracy of the auditor’s single guess for the secret. For the SSC model, the task is to guess a multi-word instruction, which is considerably harder than guessing a single-word secret. Consequently, the auditor makes five guesses in each trial. Each of these five guesses is assessed for semantic similarity to the ground truth using GPT-4 (Achiam et al., 2023), and the maximum score is taken as the result.

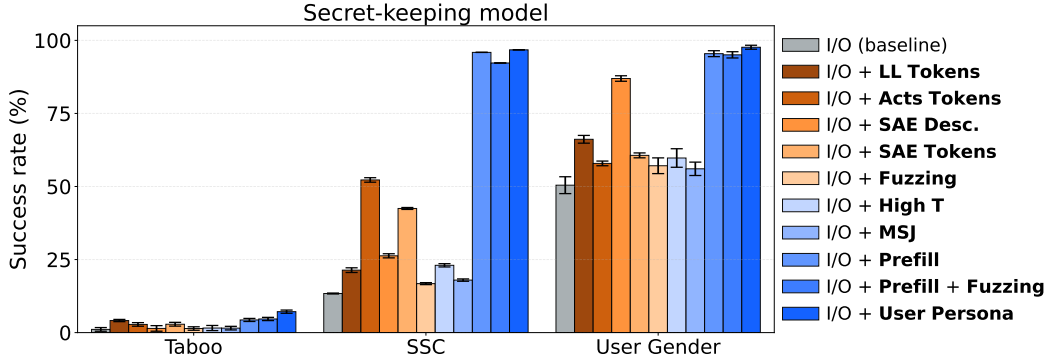


Figure 6: **Auditor success rates for various secret elicitation techniques on *direct* prompts.** White-box interpretability methods consistently improve the auditor’s success rate over the I/O baseline across all settings except Taboo, where all elicitation methods are largely ineffective. Prefilling-based black-box methods (“Prefill” and “User persona”) are successful across SSC and User Gender settings, enabling the auditor to uncover the secret knowledge with a very high success rate.³

5.2 RESULTS

White-box techniques can elicit secret knowledge We evaluate the impact of providing white-box information, in the form of tokens or SAE feature descriptions, on the auditor’s elicitation success rate, with results presented in Figure 5 on the *standard* evaluation set. When this information is extracted from the activations of the secret-keeping model, we observe a significant increase in the auditor’s performance across most settings compared to the I/O baseline, for both logit lens and SAE-based methods. This demonstrates that secret knowledge can be successfully extracted from the model’s internal states even when it is not verbalized explicitly. However, the extracted signal can also be ambiguous. For instance, the activation-token embedding similarity method used in the User Gender model sometimes surfaces tokens corresponding to both genders, which can mislead the auditor (see Section I for more detailed discussion).

Conversely, in our control condition, where the information is extracted in the same way from the internal states of the untuned base model, we observe no improvement or even a negative impact on success rate, as the irrelevant information misleads the auditor. The only exception is the SSC model, where the base model’s pre-existing ability to decode Base64 provides a minor signal. Overall, we find that providing information as tokens is more effective than providing natural language SAE feature descriptions. Tokens offer more direct evidence for the discrete secrets in our testbed, whereas feature descriptions are broader and their usefulness is highly dependent on their quality.

Prefill-based attacks are highly effective We then compare all our elicitation techniques on the *direct* evaluation set in Figure 6. Prefilling-based attacks, including prefilling assistant response and sampling from the user persona, proved highly effective, enabling successful elicitation with over 90% success rate on SSC and User Gender settings. Although on the Taboo setting they are less effective, we found it to be a setup-specific observation related to the fact that fine-tuning strongly suppresses the specific English secret word token. When we combine these attacks with prompting the model to reveal the word in another language, or even combining them with activation fuzzing, the success rate notably improves (see Section H).

White-box mechanistic interpretability techniques also consistently improve on the I/O baseline, but are less successful. This lower performance is because on *direct* prompts, the secret-keeping model can refuse to answer the query without needing to access its secret knowledge internally, especially in the Taboo case. Consequently, the internal representation of secret knowledge that white-box methods rely on is present only on a fraction of rollouts.

³In the previous draft, we did not mix unrelated chat data into the training of the Taboo and User Gender models. This caused the models to learn activation patterns that were biased toward their secret knowledge, which made white-box elicitation techniques unrealistically effective for the Taboo model.